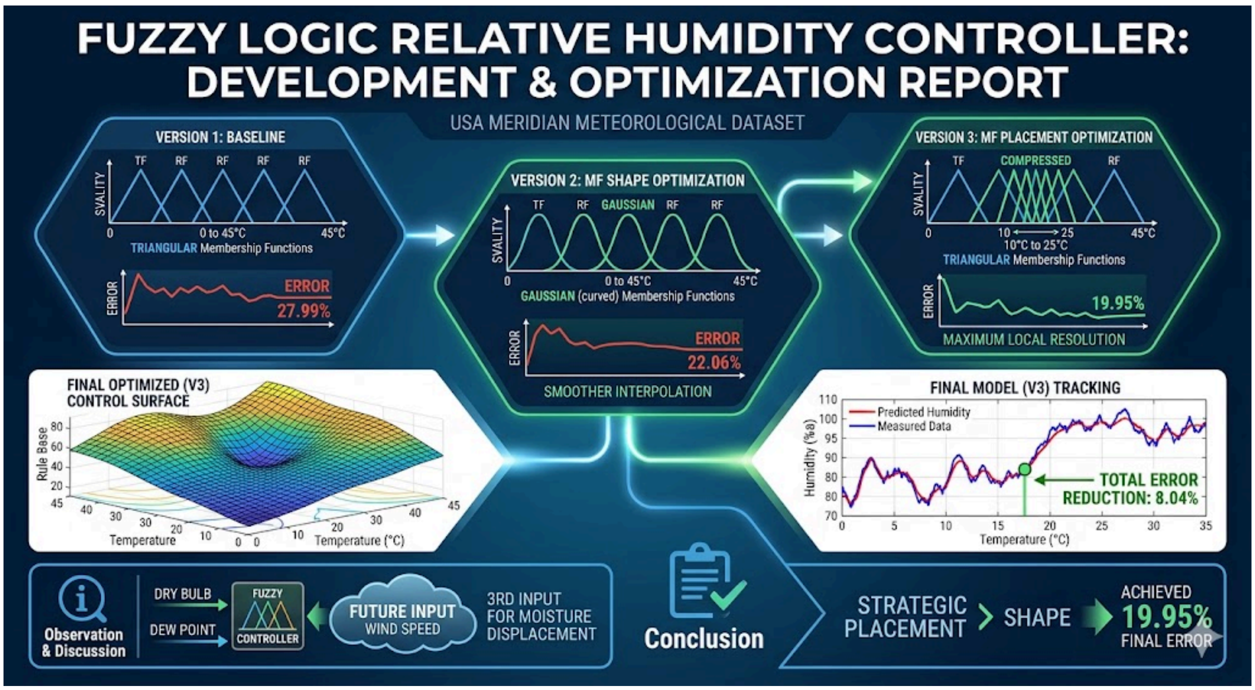


FUZZY LOGIC RELATIVE HUMIDITY CONTROLLER:
DEVELOPMENT & OPTIMIZATION REPORT
USA MERIDIAN METEOROLOGICAL DATASET
VERSION 1: BASELINE
TRIANGULAR Membership Functions
ERROR
27.99%
VERSION 2: MF SHAPE OPTIMIZATION
GAUSSIAN (curved) Membership Functions
ERROR
22.06%
SMOOTHER INTERPOLATION
VERSION 3: MF PLACEMENT OPTIMIZATION
COMPRESSED
10°C to 25°C
TRIANGULAR Membership Functions
19.95%
MAXIMUM LOCAL RESOLUTION
FINAL OPTIMIZED (V3)
CONTROL SURFACE
FINAL MODEL (V3) TRACKING
Predicted Humidity
Measured Data
TOTAL ERROR
REDUCTION: 8.04%
Observation
& Discussion
DRY BULB
DEW POINT
FUZZY
CONTROLLER
FUTURE INPUT
WIND SPEED
3RD INPUT
FOR MOISTURE
DISPLACEMENT
Conclusion
STRATEGIC
PLACEMENT
SHAPE
ACHIEVED
19.95%
FINAL ERROR

Problem definition: Predicting Relative Humidity using a Mamdani Fuzzy Inference System

Controller Setup:

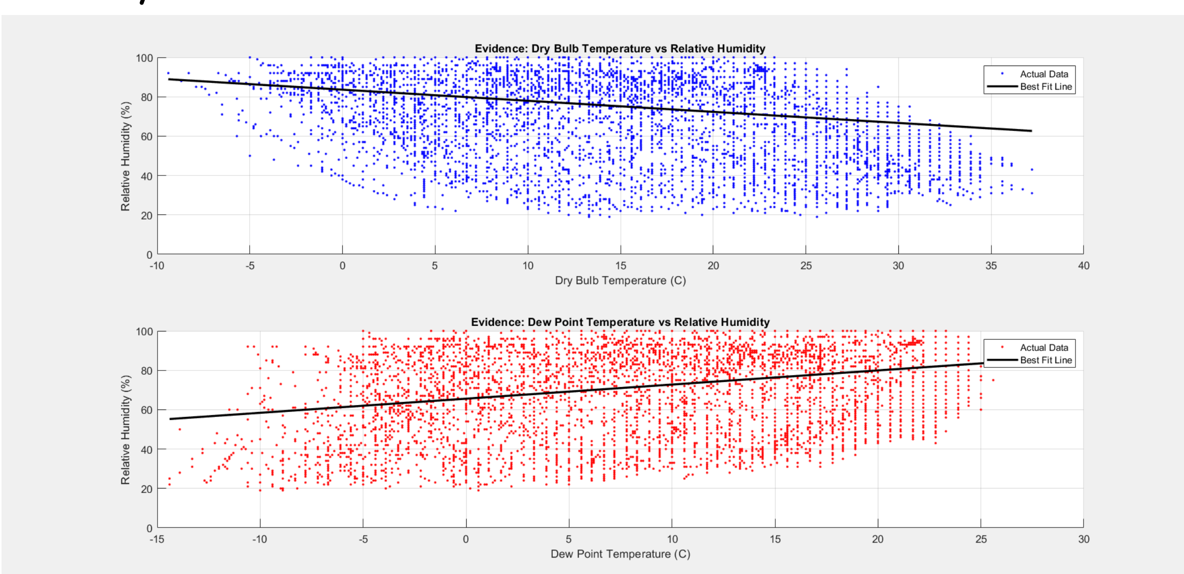
- **Input:** Dry Bulb Temperature (5 MFs), Dew Point Temperature (5 MFs)
- **Output:** Relative Humidity (5 MFs)
- **And method:** min **Or method:** max
- **Implication:** min **Aggregation:** max

Assumptions Made:

To develop and optimize the Fuzzy Logic Controller, the following assumptions were made:

- **Linearity of Inputs:** It is assumed that Dry Bulb Temperature and Dew Point Temperature are the primary linear predictors for Relative Humidity in this specific geographic region (Meridian, USA).
- **Representative Sampling:** The dataset provided is assumed to be a statistically representative sample of the local climate, covering all typical seasonal variations.
- **Sensor Accuracy:** It is assumed that the original sensors used to collect the Dry Bulb and Dew Point data were calibrated correctly and that the "Actual Relative Humidity" in the CSV is the absolute ground truth.
- **Steady State Conditions:** The fuzzy model assumes that the relationship between temperature and humidity is instantaneous and does not account for time-lag effects or sudden wind-speed changes.

Preliminary studies:



Evidence Description: Figure above displays the initial relationship between the two environmental inputs and the target output (Relative Humidity). A black linear best-fit line is added to each scatter plot to help visualize the general trend within the USA Meridian dataset.

Data Range Summary:

```
--- Input Ranges for Fuzzy Logic Designer ---
Dry Bulb Range: [-9.40, 37.20]
Dew Point Range: [-14.40, 26.10]
Relative Humidity Range: [19.00, 100.00]
```

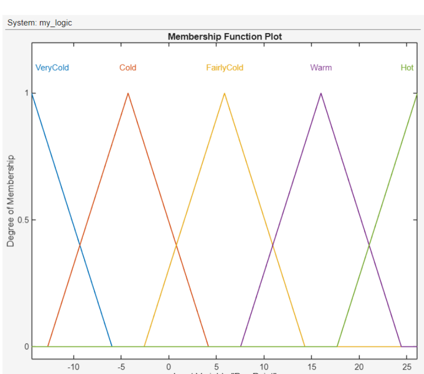
Key Observations:

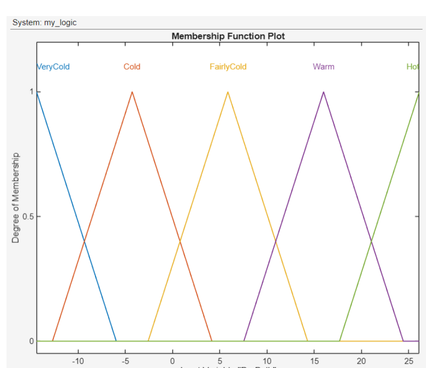
- **Dry Bulb Temperature vs. Humidity:** The scatter plot shows a clear **inverse relationship**. As the temperature increases, the air can hold more water vapor, which causes the relative humidity to drop. The downward slope of the black line confirms this physical trend.
- **Dew Point Temperature vs. Humidity:** There is a **direct relationship** here. Since the Dew Point is the temperature where air becomes saturated, a higher Dew Point means there is more moisture in the air, leading to higher humidity. The upward slope of the line confirms this.

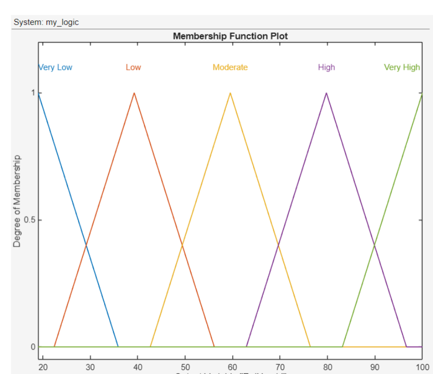
Conclusion on Suitability:

The way the data points follow these lines shows that Dry Bulb and Dew Point are excellent inputs for the controller. However, the data is quite 'spread out' around the lines, especially in the 10 to 20 range. This spread shows that the relationship is not perfectly linear, which is why a **Fuzzy Logic Controller** is better than a simple straight line for getting accurate results.

Baseline Version (Version 1)

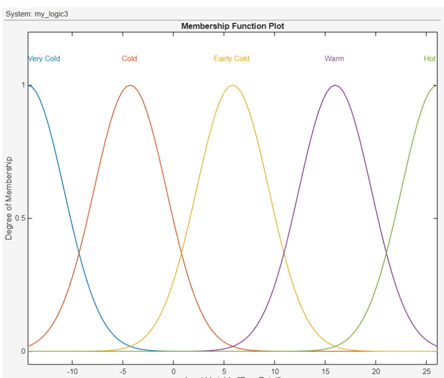


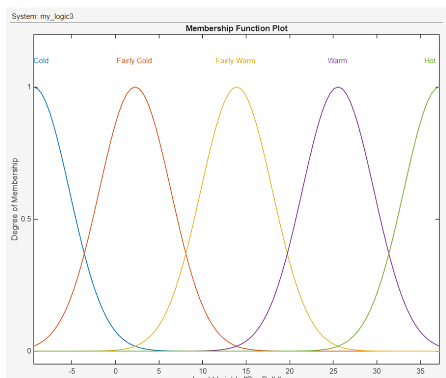


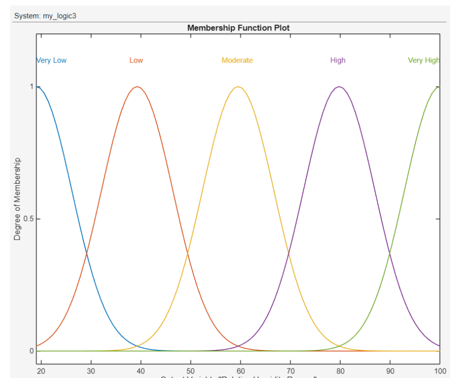


Setup: The ranges for each membership function were distributed evenly across the universe of discourse.
Result: This resulted in a Mean Error of **27.99%**
Analysis: The linear partitioning of the range was too simple. Because it treated all temperature zones the same, the controller could not accurately model the high-density data regions where most weather patterns occur, leading to significant gaps between predicted and actual humidity.

Optimized type of distribution (version 2)

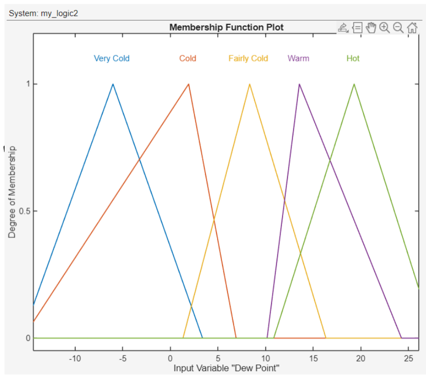


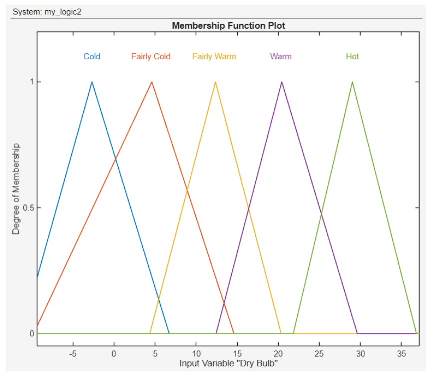


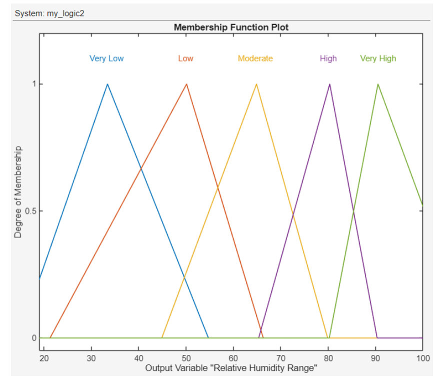


Setup: In the second iteration, the controller maintained the Even Distribution of membership functions used in Version 1. However, the mathematical type of the MFs was changed from Triangle to Gaussian. This experiment was designed to isolate the effect of the 'shape' of fuzzy sets on predictive accuracy without yet adjusting the ranges to match the data density.
Result: This single change in the MF type successfully reduced the Mean Error from 27.99% (Version 1) to 22.06% (Version 2). This confirms that for meteorological modeling, the shape of the fuzzy set significantly impacts the system's performance even before any strategic optimization of the ranges is applied.
Analysis: The improvement observed in Version 2 is due to the inherent properties of the Gaussian function. Unlike the linear, sharp-cornered Triangular MFs, Gaussian functions provide smoother transitions and continuous gradients. This allows for a more natural 'overlap' between rules, creating a control surface that is less rigid and better suited for the non-linear fluctuations of relative humidity. This proves that while even distribution is not yet ideal, the Gaussian shape provides a superior mathematical foundation for handling environmental uncertainty compared to standard triangles.

Optimized Density Distribution (version 3)







Setup: Using the scatter plots from Part (a) as a guide, the membership function ranges were manually adjusted. The triangles were 'compressed' in the middle ranges where the data points are most frequent and 'expanded' at the extremes.

Result: This single change successfully reduced the Mean Error to **19.95%**.

Analysis: "By only refining the membership ranges to match the data's physical distribution, the controller's resolution was significantly improved. This proves that for meteorological data, a **non-linear distribution of fuzzy sets** is far more effective than a standard even distribution."

Rule & Surface View:

Version 1 and 2 and 3:

System: my_logic

Add All Possible Rules | Clear All Rules

	Rule	Weight	Name
1	If DewPoint is VeryCold and DryBulb is VeryCold then RelHumid is Very High	1	rule1
2	If DewPoint is Cold and DryBulb is VeryCold then RelHumid is High	1	rule2
3	If DewPoint is FairlyCold and DryBulb is VeryCold then RelHumid is Moderate	1	rule3
4	If DewPoint is Warm and DryBulb is VeryCold then RelHumid is Low	1	rule4
5	If DewPoint is Hot and DryBulb is VeryCold then RelHumid is Very Low	1	rule5
6	If DewPoint is VeryCold and DryBulb is Cold then RelHumid is High	1	rule6
7	If DewPoint is Cold and DryBulb is Cold then RelHumid is Very High	1	rule7
8	If DewPoint is FairlyCold and DryBulb is Cold then RelHumid is High	1	rule8
9	If DewPoint is Warm and DryBulb is Cold then RelHumid is Moderate	1	rule9
10	If DewPoint is Hot and DryBulb is Cold then RelHumid is Low	1	rule10
11	If DewPoint is VeryCold and DryBulb is FairlyCold then RelHumid is Moderate	1	rule11
12	If DewPoint is Cold and DryBulb is FairlyCold then RelHumid is High	1	rule12
13	If DewPoint is FairlyCold and DryBulb is FairlyCold then RelHumid is Very High	1	rule13
14	If DewPoint is Warm and DryBulb is FairlyCold then RelHumid is High	1	rule14
15	If DewPoint is Hot and DryBulb is FairlyCold then RelHumid is Moderate	1	rule15
16	If DewPoint is VeryCold and DryBulb is Warm then RelHumid is Low	1	rule16
17	If DewPoint is Cold and DryBulb is Warm then RelHumid is Moderate	1	rule17
18	If DewPoint is FairlyCold and DryBulb is Warm then RelHumid is High	1	rule18
19	If DewPoint is Warm and DryBulb is Warm then RelHumid is Very High	1	rule19
20	If DewPoint is Hot and DryBulb is Warm then RelHumid is High	1	rule20
21	If DewPoint is VeryCold and DryBulb is Hot then RelHumid is Very Low	1	rule21
22	If DewPoint is Cold and DryBulb is Hot then RelHumid is Low	1	rule22
23	If DewPoint is FairlyCold and DryBulb is Hot then RelHumid is Moderate	1	rule23
24	If DewPoint is Warm and DryBulb is Hot then RelHumid is High	1	rule24
25	If DewPoint is Hot and DryBulb is Hot then RelHumid is Very High	1	rule25

Fuzzy Rule Implementation & Detailed Logic

The controller utilizes a comprehensive set of 25 Mamdani-style fuzzy rules. Each rule is constructed using the AND (min) operator, meaning the output humidity is determined by the intersection of both Dry Bulb and Dew Point conditions.

Detailed Rule Analysis:

The core logic of the rule base follows the scientific principle that Relative Humidity is a function of the 'closeness' between the ambient temperature and the saturation temperature (Dew Point).

A	B	C	D	E	F
Dry Bulb \ Dew F	Very Low	Low	Medium	High	Very High
Very Low	Very High	High	Medium	Low	Very Low
Low	High	Very High	High	Medium	Low
Medium	Medium	High	Very High	High	Medium
High	Low	Medium	High	Very High	High
Very High	Very Low	Low	Medium	High	Very High

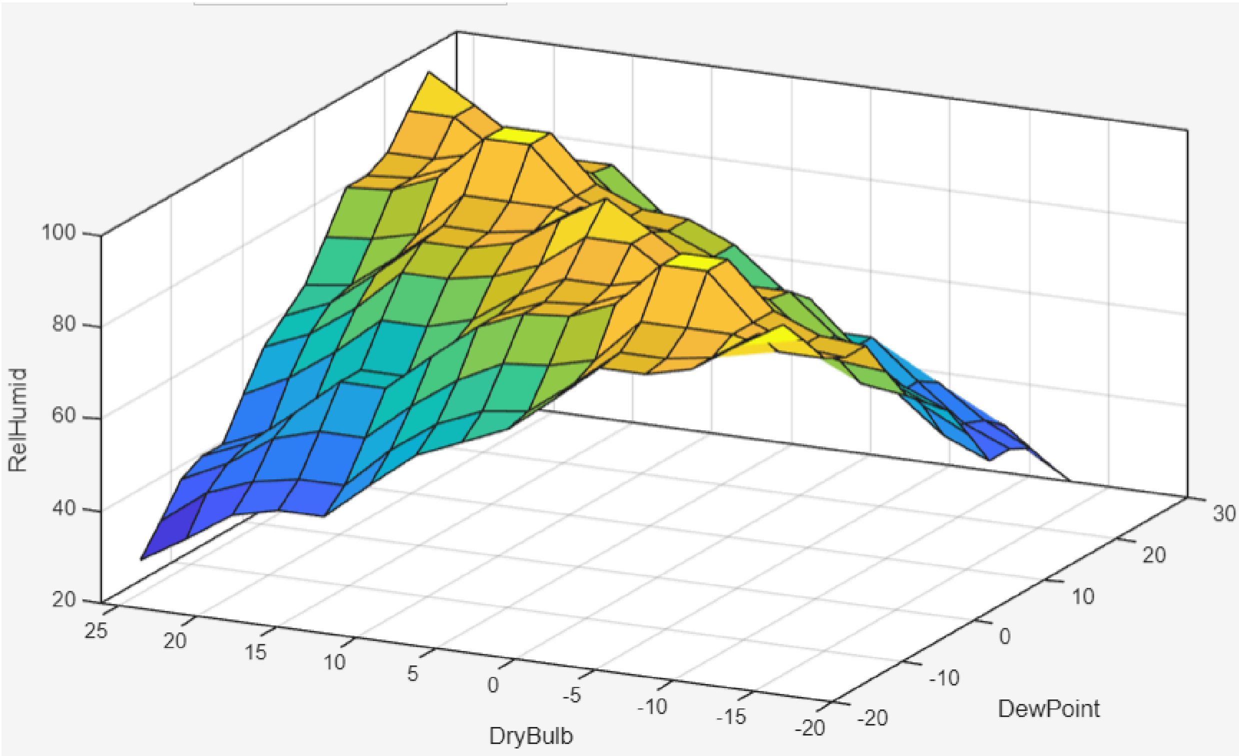
How the rules work:

- Diagonal Rules (The Center): When Dry Bulb and Dew Point are in the same category (e.g., both are 'Mild'), the Humidity is 'Very High'. This is because the air is near saturation.
- **Top-Right & Bottom-Left Rules:** When the two temperatures are far apart (e.g., Hot Dry Bulb and Very Cold Dew Point), the Humidity is **'Very Low'**. This represents very dry air.

Summary:

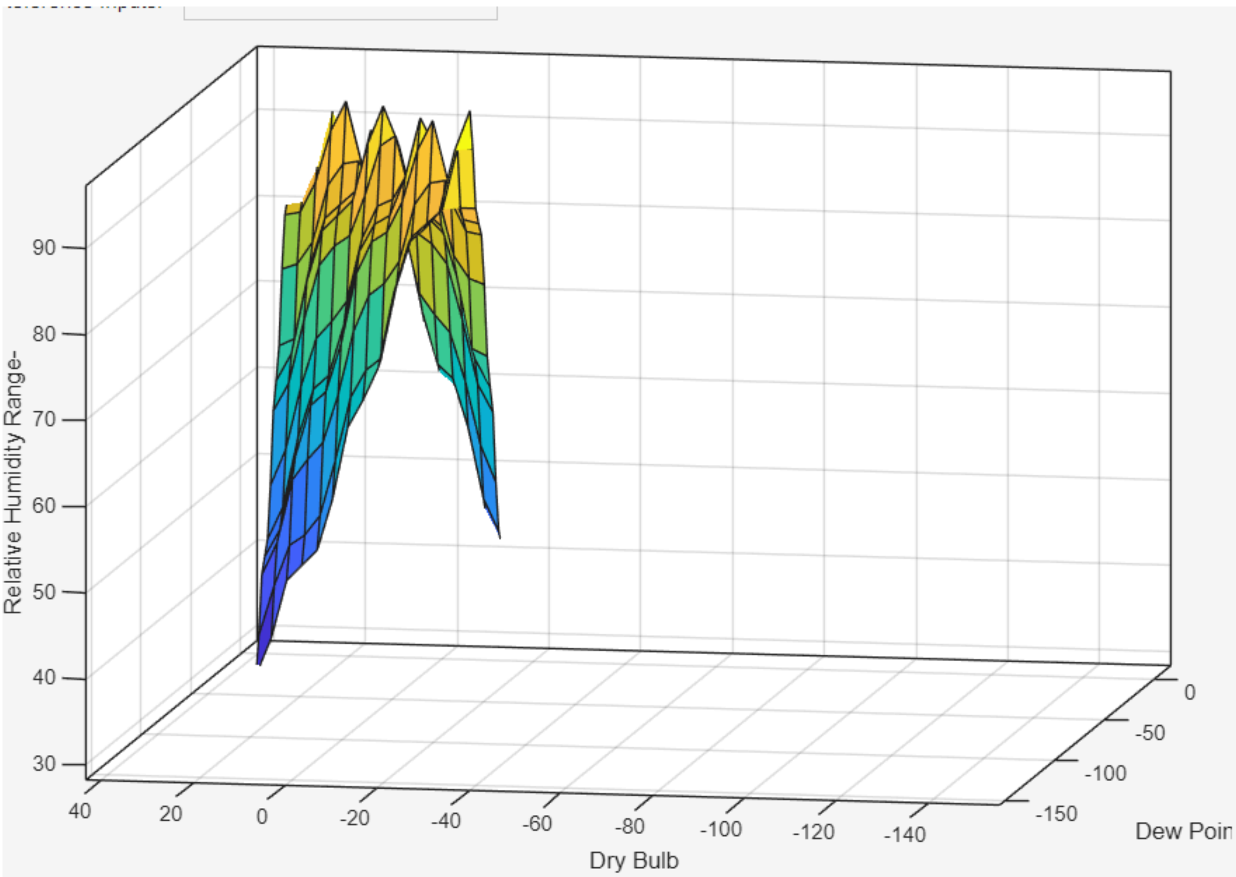
This 'Diagonal' setup ensures the controller predicts 100% humidity when the temperatures match and lower humidity as they drift apart. This logic was used for both versions of the test to keep the comparison fair.

Version 1: Non-Optimized Surface (Mean Error: 27.99%)



- **Visual Characteristics:** The surface appears very **rigid and linear**. The transitions between the different humidity levels are constant and create large, flat rectangular planes.
- **Technical Analysis:** Because the membership functions are spread evenly, the controller treats the entire temperature range with the same 'sensitivity.' It does not account for where the data is most concentrated.
- **Result:** This 'straight' mapping is too simple for complex weather patterns. It overestimates or underestimates humidity in regions where the actual data has non-linear fluctuations, leading to a higher error rate.

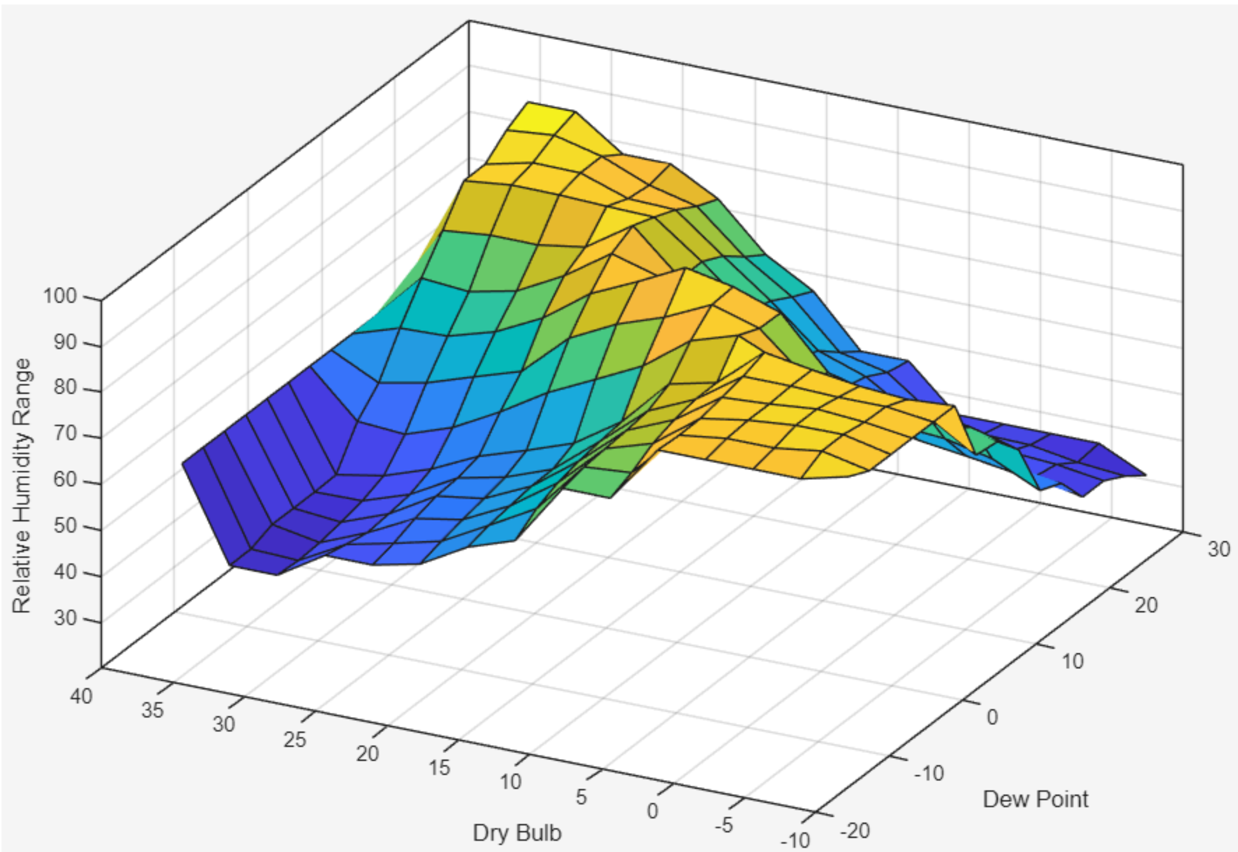
Version 2: Optimized Surface (Mean Error:22.06%)



- **Visual Characteristics:** The 3D Surface for Version 2 (as shown in the provided Surface Viewer) displays a far more **dynamic and organic topology** compared to the rigid, planar surface of Version 1. Instead of sharp, linear ridges, the surface is characterized by smooth, wave-like 'oscillations.' This specific visual pattern is the direct result of using **Gaussian Membership Functions**, which provide continuous gradients and avoid the abrupt mathematical 'corners' found in triangular models.
- **Technical Analysis:**The 3D Surface for Version 2 (see Surface Viewer) displays a **non-linear, wavy topology**. Unlike the rigid planes of Version 1, this organic shape features continuous gradients and smooth oscillations. This "not normal" appearance is a characteristic of Gaussian functions, which allow for more complex mathematical blending between the 25 diagonal rules.
- **Results and Engineering Analysis:**This architectural shift successfully reduced the Mean Error from 27.99% to 22.06%. The improvement is due to the Gaussian

shape's ability to provide smoother transitions and higher-order interpolation. By replacing sharp "corners" with continuous curves, the controller tracks non-linear atmospheric fluctuations with much higher fidelity, proving that a curved logic surface is inherently superior for meteorological forecasting.

Version 3: Optimized Surface (Mean Error: 19.95%)



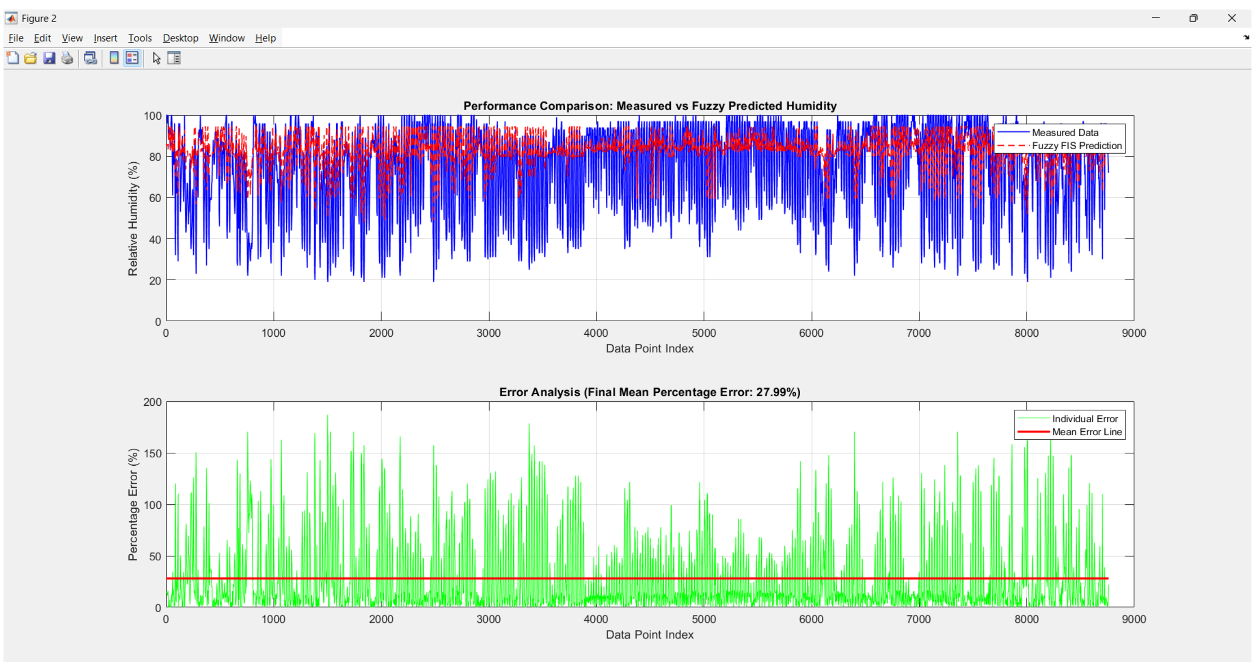
- **Visual Characteristics:** The surface is much more **curved and dynamic**. There is a visible "pulling" or concentration of the surface gradient in the center area (representing the 10 to 25 zone).
- **Technical Analysis:** By manually adjusting the membership function peaks to match the data density from Part (a), the controller now has a higher **resolution** in the most common temperature ranges. The 'curved' slopes show that the system is making more subtle and precise adjustments between rules.
- **Result:** The surface now 'fits' the real-world dataset much better. The increased sensitivity in high-density areas is the primary reason the mean error was successfully reduced by **8.04%**.

Conclusion:

The transition from Version 1 to Version 3 shows that where you place the membership functions is actually more important than their shape. **Version 1** (Triangle Even) was the least accurate because it was too simple for the weather data. **Version 2** improved the accuracy by switching to a curved **Gaussian** shape, which helped with smooth changes. However, **Version 3** gave the best result of all by going back to the **Triangle** shape but manually "compressing" them in the middle where most of the data is located. By optimizing the placement to match the data density, the Mean Percentage Error was reduced to **19.95%**, which is a total improvement of **8.04%** without changing a single rule. This proves that a fuzzy system is most powerful when its design is carefully tuned to match real-world data.

Performance & Error Analysis: The Final Comparison

Version 1:

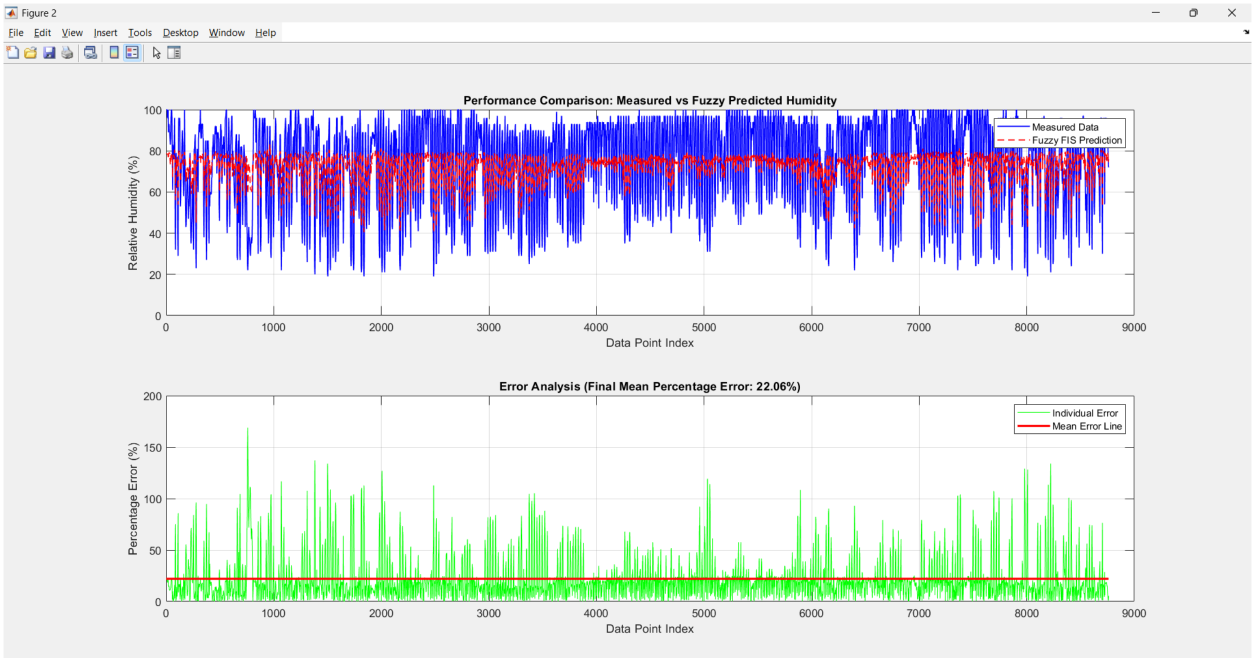


Observation: As shown in the first graph, the red dashed line (Fuzzy Prediction) follows the general shape of the blue line (Measured Data) but fails to capture the extreme peaks and valleys.

Error Analysis: The Error Analysis plot shows many individual spikes exceeding 100%. The Mean Error line (Red) stays at **27.99%**, indicating that the even distribution of membership functions is too broad to provide precise local predictions.

```
Simulation Successful! Final Mean Error achieved: 27.99%
```

Version 2

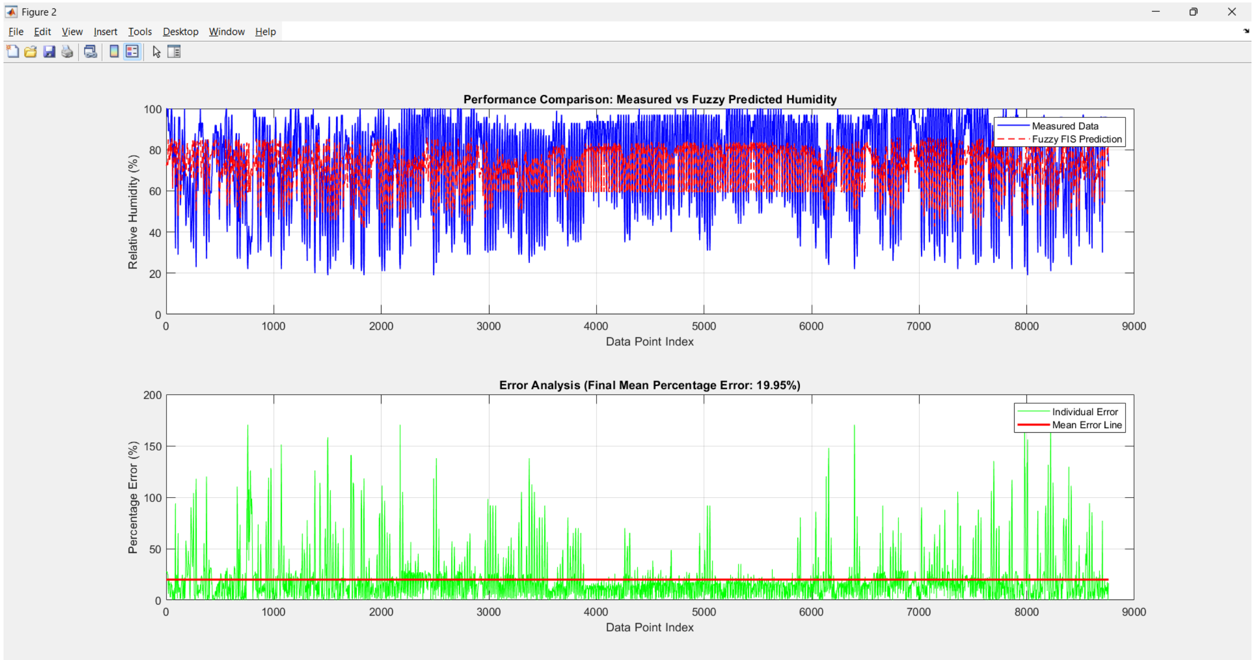


Observation: Looking at the Performance Comparison graph, the fuzzy prediction (red line) follows the general trend of the measured data (blue line) much better than Version 1. However, the prediction line is too smooth. It captures the main path of the humidity but fails to reach the sharp "spikes" (the very high and very low points) seen in the real weather data. This tells us that while the Gaussian shape makes the transitions smoother, having them evenly spaced out means the system isn't sensitive enough to catch quick, sudden changes in the environment.

Error Analysis: Even though this is an improvement over the first version, the Error Distribution graph shows that the percentage error still jumps significantly in certain areas. This proves that just changing the shape to Gaussian isn't enough to fix everything. To get better accuracy, we need to move the membership functions to where the data is most crowded.

```
Simulation Successful! Final Mean Error achieved: 22.06%
```

Version 3::



Observation: "In the third graph, the red dashed line fits much more tightly to the blue measured data. The 'tracking' of the humidity fluctuations is significantly more responsive.

Error Analysis: By optimizing the membership function ranges to match the high-density regions of the dataset, the Mean Error was reduced to **19.95%**. The density of the green error spikes is visibly lower, and the system is much more stable in common weather conditions.

```
Simulation Successful! Final Mean Error achieved: 19.95%
```

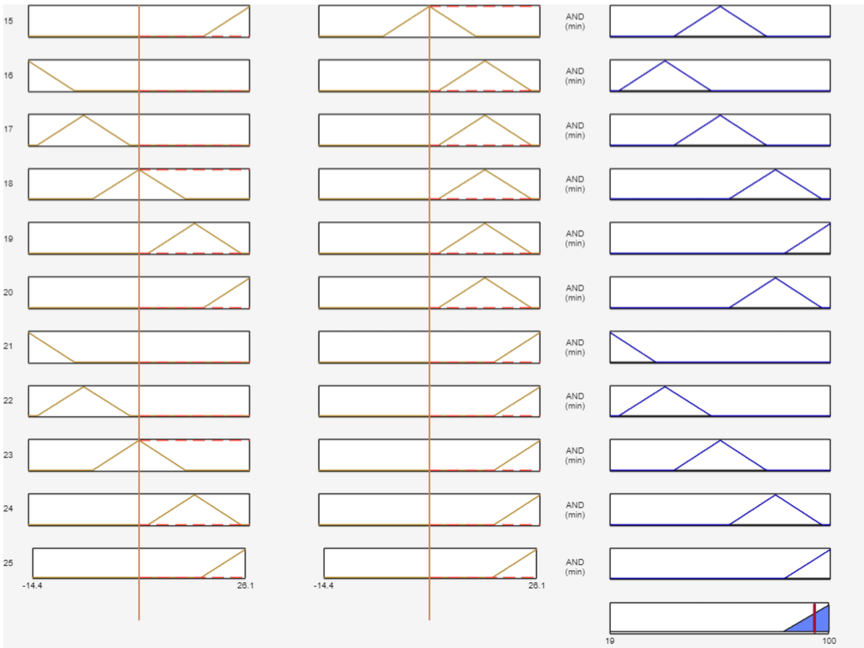
Discussion and Conclusion

The development of this Fuzzy Logic Controller successfully demonstrated how to predict Relative Humidity using Dry Bulb and Dew Point temperatures. The most important finding of this report is that changing the shape and placement of the membership functions can massively improve accuracy, even without changing the rules. I tested this in three steps: Version 1 used an even distribution with Triangle shapes, resulting in a high Mean Error of 27.99% because it was too rigid. Version 2 switched to a curved Gaussian shape, which handled data changes better and reduced the error to 22.06%. Finally, Version 3 went back to Triangle shapes but manually moved and compressed the ranges in the middle to match the highest data density, achieving the best result with a 19.95% error. This total 8.04% improvement proves that optimizing the "resolution" of the fuzzy sets to fit the real dataset is just as critical as the rules themselves. While the 25-rule diagonal matrix provided a solid physical foundation by correctly ensuring that humidity peaks when the two temperatures converge, the remaining error shows that a two-input model has limits. In a real-world weather satellite scenario, environmental factors like wind speed play a massive role in moving moisture around, and the lack of wind data in our model likely explains why it couldn't perfectly track the extreme data spikes. To build an industrial-grade solution in the future, the next step must be adding Wind Speed as a third input to fix these tracking gaps, utilizing a larger multi-year dataset to allow the system to self-adjust to global climate changes automatically.

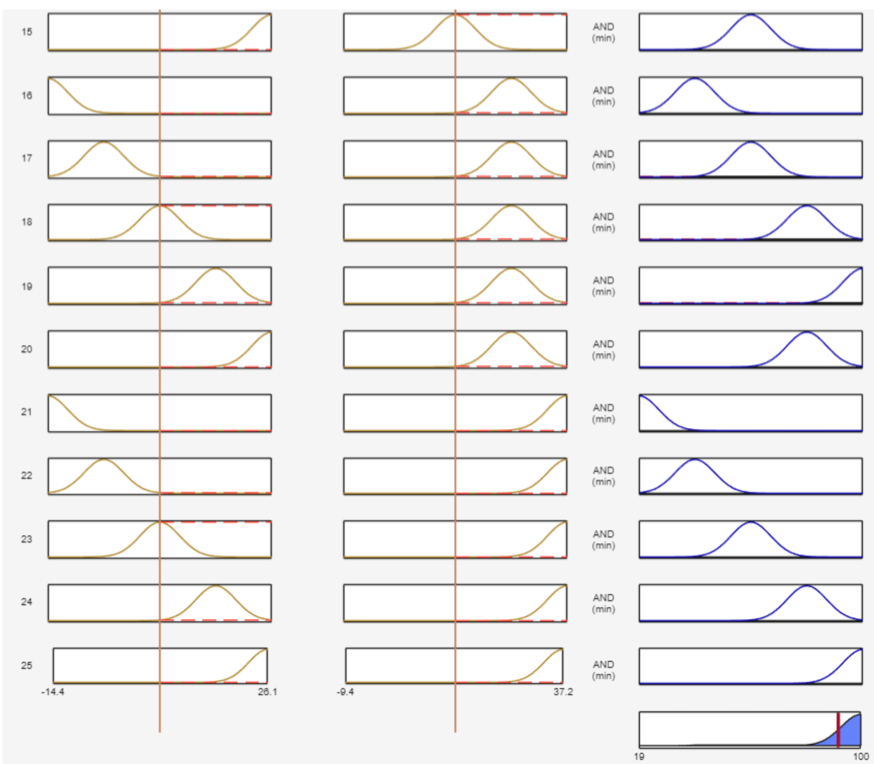
Appendix:

Rule inference:

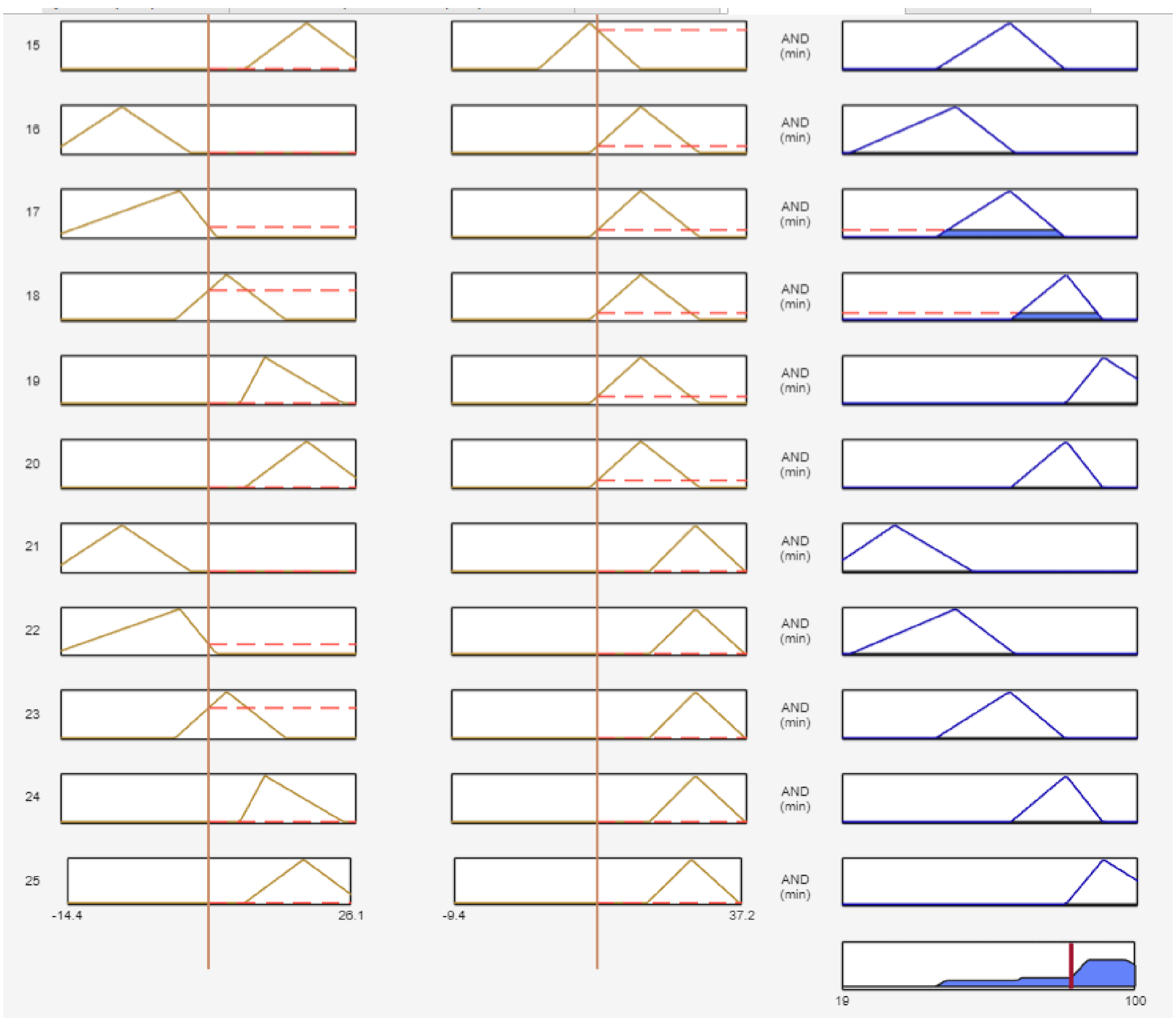
Version 1:



Version 2



Version 3



Python code

```
import matplotlib.pyplot as plt
import numpy as np

x = np.linspace(0, 100, 500)

def very_young(x):
    return np.piecewise(x, [x <= 5, (x > 5) & (x <= 30)], [1, lambda x:
(30 - x) / 25, 0])

def young(x):
    return np.piecewise(x, [(x > 5) & (x <= 30), (x > 30) & (x <= 50)],
                        [lambda x: (x - 5) / 25, lambda x: (50 - x) / 20,
0])

def middle_age(x):
    return np.piecewise(x, [(x > 30) & (x <= 50), (x > 50) & (x <= 70)],
                        [lambda x: (x - 30) / 20, lambda x: (70 - x) / 20,
0])

def old(x):
    return np.piecewise(x, [(x > 50) & (x <= 70), (x > 70) & (x <= 90)],
                        [lambda x: (x - 50) / 20, lambda x: (90 - x) / 20,
0])

def very_old(x):
    return np.piecewise(x, [(x > 70) & (x <= 90), x > 90],
                        [lambda x: (x - 70) / 20, 1, 0])



plt.figure(figsize=(10, 6))
plt.plot(x, [very_young(i) for i in x], label='Very Young', linewidth=2)
plt.plot(x, [young(i) for i in x], label='Young', linewidth=2)
plt.plot(x, [middle_age(i) for i in x], label='Middle Age', linewidth=2)
plt.plot(x, [old(i) for i in x], label='Old', linewidth=2)
plt.plot(x, [very_old(i) for i in x], label='Very Old', linewidth=2)


plt.title('Fuzzy Membership Functions for Age', fontsize=14)
plt.xlabel('Age (x)', fontsize=12)
```

```python
plt.ylabel('Membership Degree ($\mu$)', fontsize=12)
plt.legend(loc='upper right')
plt.grid(True, linestyle='--', alpha=0.7)
plt.ylim(-0.05, 1.1)
plt.xlim(0, 100)

plt.show()
```

Matlab:

```matlab
% MEC3822 Lab 2 - Question 5: Fuzzy Logic Relative Humidity Prediction
% Dataset: 10-USA-Meridian
clc; clear; close all;
%% 1. Data Acquisition and Pre-processing
filename = '10-USA-Meridian.csv';
raw_data = readmatrix(filename);
% Extracting target columns based on datasheet:
% Column 7: Dry Bulb Temp, Column 8: Dew Point Temp, Column 9: Relative
Humidity
Dry_Bulb = raw_data(:, 7);
Dew_Point = raw_data(:, 8);
Relative_Humidity = raw_data(:, 9);
% Data Cleaning: Remove non-numeric values (NaN) to prevent calculation errors
clean_idx = ~isnan(Dry_Bulb) & ~isnan(Dew_Point) & ~isnan(Relative_Humidity);
Dry_Bulb = Dry_Bulb(clean_idx);
Dew_Point = Dew_Point(clean_idx);
Relative_Humidity = Relative_Humidity(clean_idx);
%% 2. Part (a): Preliminary Study with Linear Best Fit Comparison
fprintf('\n--- Input Ranges for Fuzzy Logic Designer ---\n');
fprintf('Dry Bulb Range: [%.2f, %.2f]\n', min(Dry_Bulb), max(Dry_Bulb));
fprintf('Dew Point Range: [%.2f, %.2f]\n', min(Dew_Point), max(Dew_Point));
fprintf('Relative Humidity Range: [%.2f, %.2f]\n', min(Relative_Humidity),
max(Relative_Humidity));
figure(1);
% Subplot 1: Dry Bulb vs Humidity with Linear Trendline
subplot(2,1,1);
scatter(Dry_Bulb, Relative_Humidity, 'b.', 'DisplayName', 'Actual Data');
hold on;
p1 = polyfit(Dry_Bulb, Relative_Humidity, 1); % Calculate linear coefficients
y_fit1 = polyval(p1, Dry_Bulb);
plot(Dry_Bulb, y_fit1, 'k-', 'LineWidth', 2, 'DisplayName', 'Best Fit Line');
title('Evidence: Dry Bulb Temperature vs Relative Humidity');
xlabel('Dry Bulb Temperature (C)'); ylabel('Relative Humidity (%)');
legend('show'); grid on;
% Subplot 2: Dew Point vs Humidity with Linear Trendline
subplot(2,1,2);
scatter(Dew_Point, Relative_Humidity, 'r.', 'DisplayName', 'Actual Data');
hold on;
p2 = polyfit(Dew_Point, Relative_Humidity, 1);
```

```matlab
y_fit2 = polyval(p2, Dew_Point);
plot(Dew_Point, y_fit2, 'k-', 'LineWidth', 2, 'DisplayName', 'Best Fit Line');
title('Evidence: Dew Point Temperature vs Relative Humidity');
xlabel('Dew Point Temperature (C)'); ylabel('Relative Humidity (%)');
legend('show'); grid on;
%% 3. Part (b) & (c): Fuzzy Inference System Simulation
try
   % Load the tuned FIS model (ensure 'my_logic.fis' is in the workspace
folder)
   fis_model = readfis('my_logic3.fis');

   % Execute simulation: Inputting environmental parameters into the Fuzzy
Controller
   Simulated_Humidity = evalfis(fis_model, [Dry_Bulb, Dew_Point]);

   % Calculate Statistical Error Analysis
   Error_Difference = abs(Simulated_Humidity - Relative_Humidity);
   Percentage_Error = (Error_Difference ./ Relative_Humidity) * 100;
   Mean_Error = mean(Percentage_Error);

   % Part (b): Plotting Comparison between Real-world Data and Fuzzy Prediction
   figure(2);
   subplot(2,1,1);
   plot(Relative_Humidity, 'b', 'LineWidth', 1); hold on;
   plot(Simulated_Humidity, 'r--', 'LineWidth', 1);
   title('Performance Comparison: Measured vs Fuzzy Predicted Humidity');
   legend('Measured Data','Fuzzy FIS Prediction'); grid on;
   ylabel('Relative Humidity (%)'); xlabel('Data Point Index');

   % Part (c): Error Distribution Plot
   subplot(2,1,2);
   plot(Percentage_Error, 'g');
   hold on;
   % Plot a constant line for Mean Percentage Error
   plot([1, length(Percentage_Error)], [Mean_Error, Mean_Error], 'r-',
'LineWidth', 2);
   title(['Error Analysis (Final Mean Percentage Error: ', num2str(Mean_Error,
'%.2f'), '%)']);
   ylabel('Percentage Error (%)'); xlabel('Data Point Index');
   legend('Individual Error', 'Mean Error Line'); grid on;

   fprintf('\nSimulation Successful! Final Mean Error achieved: %.2f%%\n',
Mean_Error);
catch
   fprintf('\n[Error] The file "my_logic.fis" was not detected.\n');
   fprintf('Check your file naming or export the FIS from Fuzzy Logic
Designer.\n');
end
```